

The rank-zero theorem for $\lambda = (3, 2, 1^{n-5})$ via a three-branch reduction

Clio

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Abstract

We prove $\Pi^{S_n} |_{V_{(3,2,1^{n-5})}} = 0$ for every $n \geq 8$. This is the first family in the rank-zero programme where λ has *two* length-preserving corners, so $B_{\ell+1}^{(\lambda)} V_\lambda$ is 2-dimensional rather than 1-dimensional. Empirically the j -formula at $j = \ell+1$ does not collapse the image to dimension 1, which was the obstacle flagged in the May-19 note. We resolve it by showing the 2-dimensional image is nevertheless *entirely contained in* E_1^- , so the rightmost (T_1+1) of R'_ℓ annihilates it.

The proof is a three-branch generalization of the May-15 branching reduction. The $+q$ -eigenspace $E_{n-1}^+ |_{V_\lambda}$ decomposes, as an $H_q(S_{n-2})$ -module, into *three* pieces $\Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C})$ indexed by the three pairs of corners of λ . Two pieces (those involving the column-1 bottom corner) reduce to the known j -formula for $\mu_A = (2, 2, 1^{n-6})$ (May-15/18 family) and $\mu_B = (3, 1^{n-5})$ (May-12 hook), each contributing a 1-dimensional image in E_1^- . The third piece, indexed by $\mu_C = (2, 1^{n-4})$ (May-11 hook), contributes *zero* because its j -formula sits one step earlier, so the May-15 commutation crosses the E_1^- wall and is killed by a leading (T_1+1) .

Status. Conditional only on Phase A endpoint at λ (immediate from May-19 abstract Phase A theorem). All other ingredients are rigorously proved in prior writeups.

Verification. $B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-$ verified at $q = 7$ for $n \in \{8, 9, 10\}$ (dim 2 each, full collapse to 0 after one more R'_ℓ). Dimension decomposition $\dim V_{\mu_A} + \dim V_{\mu_B} + \dim V_{\mu_C} = \dim E_{n-1}^+ V_\lambda$ verified through $n = 11$.

1 Setup

We use the conventions of the May-13–May-19 papers. $H_q(S_n)$ is the Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$, quadratic relation $T_i^2 = (q-1)T_i + q$, and braid relations. V_λ is the seminormal cell module for $\lambda \vdash n$, with Hoefsmit basis $\{v_T\}_{T \in \text{SYT}(\lambda)}$. E_j^+ (resp. E_j^-) is the q -eigenspace (resp. -1 -eigenspace) of T_j .

Write

$$R'_k = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} = R'_j R'_{j+1} \cdots R'_n, \quad \Pi^{S_n} = B_2^{(\lambda)}.$$

Throughout, $n \geq 8$ and

$$\lambda := (3, 2, 1^{n-5}) \vdash n, \quad \ell := \ell(\lambda) = n-3, \quad j_0 := \ell+1 = n-2.$$

The j -formula target is $B_{n-2}^{(\lambda)} = R'_{n-2} R'_{n-1} R'_n$ (3 factors).

1.1 The three corners and the three inner partitions

$\lambda = (3, 2, 1^{n-5})$ has *three* removable corners:

$$c_1 = (1, 3), \quad c_2 = (2, 2), \quad c_3 = (n-3, 1).$$

Of these, c_1 and c_2 are *length-preserving* (removing them leaves a partition of length $\ell - 1$ but with the same number of rows occupied as λ ignoring trailing parts; more precisely, $\lambda_r > \lambda_{r+1}$ at $r = 1, 2$), and c_3 is the column-1 bottom corner (length-reducing). The rank-corner formula (May-11) predicts $\dim B_{\ell+1}^{(\lambda)} V_\lambda = \#\{\text{length-pres corners}\} = 2$; this matches our computation (see Section 6).

For each *pair* of corners $\{c_i, c_j\}$, the doubly-stripped partition (remove both cells) is

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1, c_3\} = (2, 2, 1^{n-6}) \vdash n-2, & \text{("A"=length-pres } c_1 \text{ paired with bottom } c_3)} \\ \mu_B &:= \lambda \setminus \{c_2, c_3\} = (3, 1^{n-5}) \vdash n-2, & \text{("B"=length-pres } c_2 \text{ paired with bottom } c_3)} \\ \mu_C &:= \lambda \setminus \{c_1, c_2\} = (2, 1^{n-4}) \vdash n-2. & \text{("C"=the two length-pres corners with each other)} \end{aligned}$$

μ_A is the second non-hook (May-15 family at $a = 2$), μ_B is the column-1 hook of length ≥ 3 in the first row (May-12), μ_C is the column-1 hook of length 2 in the first row (May-11).

1.2 Known rank-zero results we will use

Each μ_X is in its own rank-zero regime, and the rank-zero theorem is known structurally:

Theorem 1 (*j*-formula at μ_A , May-13 + May-18). *For $\mu_A = (2, 2, 1^{m-4}) \vdash m$ with $m \geq 6$, $B_{m-1}^{(\mu_A)} V_{\mu_A}$ is 1-dimensional and contained in $E_1^- |_{V_{\mu_A}}$.*

Theorem 2 (*j*-formula at μ_B , May-12 hook). *For $\mu_B = (3, 1^{m-3}) \vdash m$ with $m \geq 6$, $B_{m-1}^{(\mu_B)} V_{\mu_B}$ is 1-dimensional and contained in $E_1^- |_{V_{\mu_B}}$.*

Theorem 3 (*j*-formula at μ_C , May-11 hook). *For $\mu_C = (2, 1^{m-2}) \vdash m$ with $m \geq 4$, $B_m^{(\mu_C)} V_{\mu_C} = R_m^{(\mu_C)} V_{\mu_C}$ is 1-dimensional and contained in $E_1^- |_{V_{\mu_C}}$.*

Theorem 4 (Abstract Phase A endpoint, May-19). *For any $H_q(S_n)$ -module V and any $1 \leq j \leq n-1$, $W_j(V) := (T_j+1) \cdots (T_1+1)V = E_j^+(V)$. In particular, $R'_n V = E_{n-1}^+(V)$.*

We apply Theorem 4 at $V = V_\lambda$ and at $V = V_{\mu_X}$ ($X \in \{A, B, C\}$); we apply Theorems 1–3 at $m = n-2$, noting that $m \geq 6$ for $n \geq 8$.

2 The branching isomorphisms Φ_X

We mimic the May-15 construction for each pair of corners. Recall that $H_q(S_{n-2})$ embeds in $H_q(S_n)$ as the subalgebra generated by $\{T_1, \dots, T_{n-3}\}$, and T_{n-1} commutes with every T_i for $i \leq n-3$ (index gap ≥ 2). Hence $E_{n-1}^+ |_{V_\lambda}$ is an $H_q(S_{n-2})$ -submodule of V_λ .

Definition 5 (Φ_X , $X \in \{A, B, C\}$). Fix $X \in \{A, B, C\}$ and write $\{c_X^-, c_X^+\} = \{c_i, c_j\}$ as the ordered pair (by row, lower-row first) corresponding to μ_X :

$$\begin{aligned} X = A : (c_A^-, c_A^+) &= (c_1, c_3) = ((1, 3), (n-3, 1)); & X = B : (c_B^-, c_B^+) &= ((2, 2), (n-3, 1)); \\ X = C : (c_C^-, c_C^+) &= (c_1, c_2) = ((1, 3), (2, 2)). \end{aligned}$$

For each SYT T' of μ_X on letters $\{1, \dots, n-2\}$, define two SYTs T_A^X, T_B^X of λ on letters $\{1, \dots, n\}$ by

- T_A^X : extend T' by placing $n-1$ at c_X^- and n at c_X^+ .
- T_B^X : extend T' by placing n at c_X^- and $n-1$ at c_X^+ .

Both are valid SYTs of λ (since $n-1, n$ exceed all other entries, the column- and row-increase conditions involving the added cells are automatic; the cells $c_X^\pm$ are corners of λ once T' has been placed).

The axial distance $d_X := c(c_X^+) - c(c_X^-)$ (content difference) is non-zero of absolute value ≥ 2 in all three cases:

$$d_A = (1-(n-3))-(3-1) = 4-n, \quad d_B = (1-(n-3))-(2-2) = 4-n, \quad d_C = (2-2)-(3-1) = -2.$$

For $n \geq 8$, $|d_A| = |d_B| = n-4 \geq 4$ and $|d_C| = 2$, so all T_{n-1} -2-blocks $\{v_{T_A^X}, v_{T_B^X}\}$ are non-degenerate over $\mathbb{Q}(q)$ (the Hoefsmit ratio $\tau(d) \in \mathbb{Q}(q)^\times$ for $|d| \geq 2$).

Define

$$\Phi_X(v_{T'}^{(\mu_X)}) := \text{the } T_{n-1} + q\text{-eigenvector in the 2-block } \{v_{T_A^X}, v_{T_B^X}\} \subset V_\lambda,$$

normalized so the coefficient of $v_{T_A^X}$ equals 1. Extend linearly to $\Phi_X : V_{\mu_X} \rightarrow V_\lambda$.

Lemma 6 (Equivariance of Φ_X). *For every $X \in \{A, B, C\}$, every $1 \leq i \leq n-3$ and every $v \in V_{\mu_X}$, $T_i \Phi_X(v) = \Phi_X(T_i v)$.*

Proof. By linearity, suffices to check $v = v_{T'}^{(\mu_X)}$. The positions of $i, i+1$ in T_A^X and T_B^X are identical to the positions in T' (since $i, i+1 \leq n-2$ and the cells of T_A^X, T_B^X holding entries $\leq n-2$ agree with T'). Three cases for the T_i -action on T' (the argument is verbatim May-15 Lemma 3.2 in our notation):

(a) **Same row in T' .** $T_i v_{T'}^{(\mu_X)} = q v_{T'}^{(\mu_X)}$. Since the positions of $i, i+1$ in T_A^X, T_B^X are the same as in T' , both $T_i v_{T_A^X} = q v_{T_A^X}$ and $T_i v_{T_B^X} = q v_{T_B^X}$, hence $T_i \Phi_X(v_{T'}^{(\mu_X)}) = q \Phi_X(v_{T'}^{(\mu_X)})$.

(b) **Same column in T' .** Symmetric to (a), eigenvalue -1 .

(c) **T_i 2-block on V_{μ_X} .** The block partner is $s_i T'$ with axial distance d_i . T_i commutes with T_{n-1} , so the 4-dim subspace $\text{span}(v_{T_A^X}, v_{T_B^X}, v_{s_i T_A^X}, v_{s_i T_B^X})$ is preserved by both, and the actions decouple:

- T_{n-1} : 2-blocks $\{v_{T_A^X}, v_{T_B^X}\}$ and $\{v_{s_i T_A^X}, v_{s_i T_B^X}\}$, axial distance d_X (same in both).
- T_i : 2-blocks $\{v_{T_A^X}, v_{s_i T_A^X}\}$ and $\{v_{T_B^X}, v_{s_i T_B^X}\}$, axial distance d_i (same in both).

The Hoefsmit coefficients on each T_{n-1} -2-block are determined by d_X alone, so the same $\tau_X := \tau(d_X)$ appears in both block-partner relations. Similarly the same a_i, b_i appear in $T_i v_{T_A^X} = a_i v_{T_A^X} + b_i v_{s_i T_A^X}$ and $T_i v_{T_B^X} = a_i v_{T_B^X} + b_i v_{s_i T_B^X}$ (because the axial distance d_i is the same). Hence:

$$\begin{aligned} T_i \Phi_X(v_{T'}^{(\mu_X)}) &= T_i(v_{T_A^X} + \tau_X v_{T_B^X}) = a_i v_{T_A^X} + b_i v_{s_i T_A^X} + \tau_X(a_i v_{T_B^X} + b_i v_{s_i T_B^X}) \\ &= a_i(v_{T_A^X} + \tau_X v_{T_B^X}) + b_i(v_{s_i T_A^X} + \tau_X v_{s_i T_B^X}) \\ &= a_i \Phi_X(v_{T'}^{(\mu_X)}) + b_i \Phi_X(v_{s_i T'}^{(\mu_X)}) = \Phi_X(T_i v_{T'}^{(\mu_X)}). \end{aligned} \quad \square$$

Lemma 7 (Φ_X is injective). *For each $X \in \{A, B, C\}$, $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ is injective. Moreover, the seminormal supports of $\Phi_A(V_{\mu_A}), \Phi_B(V_{\mu_B}), \Phi_C(V_{\mu_C})$ are pairwise disjoint, so the sum*

$$\Phi_A(V_{\mu_A}) + \Phi_B(V_{\mu_B}) + \Phi_C(V_{\mu_C})$$

is direct.

Proof. For injectivity, observe that distinct T', T'' of μ_X give disjoint pairs $\{T_A^X, T_B^X\}$ (the cells of $n-1, n$ are fixed at $c_X^\pm$, so μ_X is recovered by deletion). Hence $\{\Phi_X(v_{T'}^{(\mu_X)})\}_{T' \in \text{SYT}(\mu_X)}$ is linearly independent in V_λ (each is supported on a unique disjoint 2-element subset of the Hoefsmit basis of V_λ).

For pairwise disjointness across X : $\Phi_X(v_{T'}^{(\mu_X)})$ is supported on SYTs with $\{n-1, n\}$ at the cell pair $\{c_X^-, c_X^+\}$. Across $X \in \{A, B, C\}$, these cell pairs are distinct, so the supports are pairwise disjoint. $\square$

Lemma 8 (Image equals $E_{n-1}^+|_{V_\lambda}$). $\Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) = E_{n-1}^+|_{V_\lambda}$.

Proof. Containment in E_{n-1}^+ . By construction each $\Phi_X(v_{T'}^{(\mu_X)})$ is a $T_{n-1} + q$ -eigenvector.

Equality by dimension count. Every $T_{n-1} + q$ -eigenvector in the seminormal basis of V_λ comes from a (necessarily non-degenerate) 2-block where $\{n-1, n\}$ occupy two different corners of λ at different rows and columns. The three corner pairs are $\{c_1, c_3\} = A$, $\{c_2, c_3\} = B$, $\{c_1, c_2\} = C$ (all in different rows/cols). The number of SYTs whose $\{n-1, n\}$ occupy a fixed corner pair $\{c^-, c^+\}$ equals $2 \cdot \#\text{SYT}(\lambda \setminus \{c^-, c^+\})$, contributing $\#\text{SYT}(\mu_X) = \dim V_{\mu_X}$ to the $+q$ -eigenspace (one $+q$ -eigenvector per 2-block). Summing,

$$\dim E_{n-1}^+|_{V_\lambda} = \dim V_{\mu_A} + \dim V_{\mu_B} + \dim V_{\mu_C}.$$

By Lemma 7 the direct sum on the LHS of the statement has the same dimension and is contained in $E_{n-1}^+|_{V_\lambda}$, hence equals it. $\square$

Remark 9. At $n = 8$: $\dim V_{\mu_A}|_6 + \dim V_{\mu_B}|_6 + \dim V_{\mu_C}|_6 = 9 + 10 + 5 = 24 = \dim E_{n-1}^+|_{V_\lambda}$ (computed at $q = 7$ in Section 6). Verified through $n = 11$ ($27 + 28 + 8 = 63$).

Lemma 10 (Φ_X preserves E_1^-). For each X , $\Phi_X(E_1^-|_{V_{\mu_X}}) \subseteq E_1^-|_{V_\lambda}$.

Proof. T_1 is in $H_q(S_{n-2})$ and Φ_X is T_1 -equivariant (Lemma 6). Hence $T_1 v = -v$ implies $T_1 \Phi_X(v) = -\Phi_X(v)$. $\square$

3 The reduction formula

By Theorem 4 applied at $V = V_\lambda$, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$, and by Lemma 8 this equals $\bigoplus_X \Phi_X(V_{\mu_X})$.

Theorem 11 (Three-branch reduction).

$$B_{n-2}^{(\lambda)} V_\lambda = (T_{n-3}+1)(T_{n-2}+1) \left(\Phi_A(B_{n-3}^{(\mu_A)} V_{\mu_A}) + \Phi_B(B_{n-3}^{(\mu_B)} V_{\mu_B}) + \Phi_C(B_{n-3}^{(\mu_C)} V_{\mu_C}) \right). \quad (1)$$

Proof. We follow the May-15 argument verbatim, applied to each Φ_X piece simultaneously.

Step 1. $R'_n V_\lambda = \bigoplus_X \Phi_X(V_{\mu_X})$, proved above.

Step 2. $R'_{n-1} = (T_{n-2}+1)R'_{n-2}^{[n-1]}$ where $R'_{n-2}^{[n-1]} = (T_{n-3}+1) \cdots (T_1+1) \in H_q(S_{n-2})$. $R'_{n-2}^{[n-1]}$ preserves $E_{n-1}^+|_{V_\lambda}$ (since $T_1, \dots, T_{n-3}$ commute with T_{n-1}). By Lemma 6 (one Φ_X at a time),

$$R'_{n-2}^{[n-1]} \Phi_X(V_{\mu_X}) = \Phi_X(R'_{n-2}^{(\mu_X)} V_{\mu_X}), \quad X \in \{A, B, C\}.$$

Hence

$$R'_{n-1} R'_n V_\lambda = (T_{n-2}+1) \sum_X \Phi_X(R'_{n-2}^{(\mu_X)} V_{\mu_X}).$$

Step 3. Apply R'_{n-2} . Write $R'_{n-2} = (T_{n-3}+1)S$ where $S = (T_{n-4}+1)\cdots(T_1+1) \in H_q(S_{n-3})$. Then S commutes with T_{n-2} (index gap ≥ 2), so $S(T_{n-2}+1) = (T_{n-2}+1)S$, giving

$$R'_{n-2}(T_{n-2}+1) = (T_{n-3}+1)S(T_{n-2}+1) = (T_{n-3}+1)(T_{n-2}+1)S.$$

$S \in H_q(S_{n-3}) \subset H_q(S_{n-2})$, so by Lemma 6, $S\Phi_X(v) = \Phi_X(R'_{n-3}^{(\mu_X)}v)$ for each X (the operator S acts on V_{μ_X} as $R'_{n-3}^{(\mu_X)}$), which uses generators $T_1, \dots, T_{n-4}$. Hence

$$\begin{aligned} B_{n-2}^{(\lambda)}V_\lambda &= R'_{n-2}(T_{n-2}+1) \sum_X \Phi_X(R'_{n-2}^{(\mu_X)}V_{\mu_X}) \\ &= (T_{n-3}+1)(T_{n-2}+1)S \sum_X \Phi_X(R'_{n-2}^{(\mu_X)}V_{\mu_X}) \\ &= (T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(R'_{n-3}^{(\mu_X)}R'_{n-2}^{(\mu_X)}V_{\mu_X}) \\ &= (T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(B_{n-3}^{(\mu_X)}V_{\mu_X}), \end{aligned}$$

which is (1). □

4 The C -piece is zero

Lemma 12 (The C -piece vanishes). $B_{n-3}^{(\mu_C)}V_{\mu_C} = 0$.

Proof. $\mu_C = (2, 1^{n-4})$ has $\ell(\mu_C) = (n-2)-1 = n-3$, so $j(\mu_C) = \ell(\mu_C)+1 = n-2$. By Theorem 3, $B_{n-2}^{(\mu_C)}V_{\mu_C} = R'_{n-2}^{(\mu_C)}V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$.

Now $B_{n-3}^{(\mu_C)}V_{\mu_C} = R'_{n-3}^{(\mu_C)}B_{n-2}^{(\mu_C)}V_{\mu_C}$. The rightmost factor of $R'_{n-3}^{(\mu_C)}$ is (T_1+1) , which annihilates $E_1^-|_{V_{\mu_C}}$: $(T_1+1)v = 0$ for $v \in E_1^-$ (since $T_1v = -v$). Hence $R'_{n-3}^{(\mu_C)}B_{n-2}^{(\mu_C)}V_{\mu_C} = 0$. □

Remark 13. Geometrically, the third corner pair $\{c_1, c_2\} = \{(1, 3), (2, 2)\}$ strips two cells from λ without consuming the column-1 bottom, leaving a hook μ_C shorter by one row than μ_A, μ_B . The shorter μ_C has its j -formula at $j(\mu_C) = n-2$ rather than $n-3$, so the index $n-3$ in our reduction sits below the j -formula and is annihilated by the leading (T_1+1) . This is the structural reason the third branch contributes nothing.

5 Main theorem

Theorem 14 (Rank-zero for $\lambda = (3, 2, 1^{n-5})$). For every $n \geq 8$,

$$B_{n-2}^{(\lambda)}V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Consequently, $\Pi^{S_n}|_{V_\lambda} = 0$.

Proof. Combine the reduction (Theorem 11) with the j -formulas for μ_A, μ_B, μ_C :

A-piece. $\mu_A = (2, 2, 1^{n-6}) \vdash n-2$ with $\ell(\mu_A) = n-4$. By Theorem 1 (applied at $m = n-2$, $\ell+1 = n-3$), $B_{n-3}^{(\mu_A)}V_{\mu_A} \subseteq E_1^-|_{V_{\mu_A}}$.

B-piece. $\mu_B = (3, 1^{n-5}) \vdash n-2$ with $\ell(\mu_B) = n-4$. By Theorem 2, $B_{n-3}^{(\mu_B)}V_{\mu_B} \subseteq E_1^-|_{V_{\mu_B}}$.

C-piece. By Lemma 12, $B_{n-3}^{(\mu_C)}V_{\mu_C} = 0$.

By Lemma 10, $\Phi_X(E_1^-|_{V_{\mu_X}}) \subseteq E_1^-|_{V_\lambda}$ for $X = A, B$ (the C case is trivial since the input is 0). Hence

$$\sum_X \Phi_X(B_{n-3}^{(\mu_X)} V_{\mu_X}) \subseteq E_1^-|_{V_\lambda}.$$

Finally, $(T_{n-3}+1)$ and $(T_{n-2}+1)$ both commute with T_1 (index gap ≥ 2 since $n-3 \geq 5, n-2 \geq 6$ for $n \geq 8$). Hence each preserves $E_1^-|_{V_\lambda}$. Applying both: $B_{n-2}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$.

For the second statement, factor $\Pi^{S_n} = R'_2 R'_3 \cdots R'_{n-3} \cdot B_{n-2}^{(\lambda)}$. The rightmost factor of R'_{n-3} is (T_1+1) , which annihilates E_1^- . Hence $\Pi^{S_n} V_\lambda = 0$. $\square$

Remark 15 (On the dimension). Theorem 14 does *not* assert $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$ — only that the image lies in E_1^- , which is all we need for rank-zero. The dimension is empirically 2 (matching the rank-corner formula $\#\{\text{length-pres corners}\} = 2$), and the proof framework gives an upper bound of $\dim \Phi_A(B_{n-3}^{(\mu_A)} V_{\mu_A}) + \dim \Phi_B(B_{n-3}^{(\mu_B)} V_{\mu_B}) \leq 1 + 1 = 2$, but lower bound ($\neq 0$ for each piece after $(T_{n-3}+1)(T_{n-2}+1)$) would require a Hoefsmits case analysis we leave for a future writeup. The structural significance is the E_1^- containment.

6 Computational verification

All checks at $q = 7$ in $\mathbb{Q}$ (rational arithmetic, no floating point).

Dimension decomposition. For each $n \in \{8, 9, 10, 11\}$, $\dim V_{\mu_A} + \dim V_{\mu_B} + \dim V_{\mu_C}$ equals $\dim E_{n-1}^+|_{V_\lambda} = \dim R'_n V_\lambda$:

n	$\dim V_\lambda$	$\dim V_{\mu_A}$	$\dim V_{\mu_B}$	$\dim V_{\mu_C}$	sum	$\dim R'_n V_\lambda$
8	64	9	10	5	24	24
9	105	14	15	6	35	35
10	160	20	21	7	48	48
11	231	27	28	8	63	63

E_1^- containment at $j = \ell + 1$. For each $n \in \{8, 9, 10\}$, $B_{n-2}^{(\lambda)} V_\lambda$ has dimension 2 and $(T_1+1) \cdot B_{n-2}^{(\lambda)} V_\lambda = 0$ (equivalent to $B_{n-2}^{(\lambda)} V_\lambda \subseteq E_1^-$):

n	$\dim R'_n V_\lambda$	$\dim R'_{n-1} R'_n V_\lambda$	$\dim B_{n-2} V_\lambda$	in E_1^- ?
8	24	8	2	yes
9	35	10	2	yes
10	48	12	2	yes

(At $n = 11$ the rank-8 step in $R'_{n-1} R'_n V_\lambda$ would land at 14; we verified $B_{n-2} V_\lambda$ has dim 2 in E_1^- via prior data in 2026-05-08-conj-81-test/out.321111.n11.log.)

Final collapse. At every tested n , one more $R'_\ell = R'_{n-3}$ application drops the rank to 0, confirming $\Pi^{S_n}|_{V_\lambda} = 0$.

Scripts: `~/projects/scratch/2026-05-20-321-family/extend_j.py`, `verify_decomp.py`.

7 Honest assessment

What is rigorously proved.

1. Theorem 11: the three-branch reduction formula, conditional on Phase A at λ (Theorem 4, abstract May-19 result, fully unconditional).
2. Theorem 14: rank-zero theorem for $\lambda = (3, 2, 1^{n-5})$ at every $n \geq 8$, unconditional once we cite Theorems 1–4 (all rigorously established in prior writeups).

What is sketched but not proved. The dimension claim $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$ (matching the rank-corner formula). We prove ≤ 2 from the reduction; a non-vanishing argument for each of the A - and B -pieces after $(T_{n-3}+1)(T_{n-2}+1)$ application would close this. This is a finite Hoefsmit case analysis analogous to May-15 Section 5; we do not carry it out here because the E_1^- containment (which is what the rank-zero theorem requires) is already proved.

Structural significance. This is the first rank-zero proof where $B_{\ell+1}^{(\lambda)} V_\lambda$ is *not* 1-dimensional. The two length-preserving corners of λ each contribute an independent 1-dimensional image in E_1^- , giving a 2-dimensional total that nonetheless lies in E_1^- . This generalizes the rank-corner formula *mechanism*: each length-preserving corner contributes one “L/R-style” choice, with the inner μ_X providing the support structure. The structural prediction is that for general λ with r length-preserving corners (rank-corner formula gives $\dim B_{\ell+1}^{(\lambda)} V_\lambda = r$), the image decomposes as a sum of r pieces, each arising from the j -formula at a doubly-stripped μ_X . Pairs of corners whose “column-1 bottom” is unavailable (length-pres + length-pres pairings) contribute 0 by the same E_1^- -leakage mechanism we used for μ_C here.

Next steps.

- (1) Prove the dim-2 lower bound for $(3, 2, 1^{n-5})$ (the missing piece in Remark above).
- (2) Test the general r -corner conjecture on the next family with $r \geq 2$: $(4, 2, 1^{n-6})$ or $(3, 3, 1^{n-6})$ (both have 2 length-preserving corners; $(4, 2, 1^{n-6})$ should give dim 2, $(3, 3, 1^{n-6})$ should give dim 1 since $\lambda_1 = \lambda_2$ so $r = 1$).
- (3) The rank-corner formula itself — still conjectural for general rank-zero λ — gains substantial evidence from this proof, since it predicts a clean structural answer (number of branches = rank).

What this closes in the rank-zero programme. After May-19 (the $(2^a, 1^{n-2a})$ family is unconditional), the rank-zero theorem holds structurally for:

- All hook families $(k, 1^{n-k})$ in rank-zero regime (May-11, May-12).
- All $(2^a, 1^{n-2a})$ families at $n \geq 3a$ (May-13 through May-19).
- Today: $(3, 2, 1^{n-5})$ at $n \geq 8$ (first *two*-corner family).

The next natural targets, in rank-zero regime: $(4, 2, 1^{n-6})$, $(3, 3, 1^{n-6})$, $(4, 3, 1^{n-7})$, $(3, 2, 2, 1^{n-7})$, and so on. The three-branch reduction framework here generalizes to each: identify the length-preserving corners, set up Φ_X for each pair $\{c_X^-, c_X^+\}$ including c_3 =bottom-column-1, and apply the relevant j -formula at μ_X .